Analytical Cumulative Examination

December 1, 1990

(55pts) 1. From distribution studies X and Y are known to have partition coefficients between water and hexane of 5.81 and 6.15 ($K=[M]_{HOH}/[M]_{hex}$). The two species are to be separated by elution with hexane in a column packed with silica gel containing adsorbed water. The ratio V_S/V_M for the packing material is 0.425.

a. Calculate the capacity factor for each of the solutes.

b. Calculate the selectivity factor.

c. How many theoretical plates will be needed to provide a resolution of 1.5?

d. How long a column is needed if the plate height of the packing is 2.2×10^{-3} cm?

e. If a flow rate of 7.10 cm/min is employed, what time will be required to elute the two species?

Name	Calculation of Derived Quantities
Linear mobile-phase velocity	$u = L/t_M$
Volume of mobile phase	$V_M = t_M F$
Capacity factor	$k' = (t_R - t_M)/t_M$
Partition coefficient	$K = \frac{k' V_M}{V_S}$
Selectivity factor	$\alpha = \frac{(t_R)_B - t_M}{(t_R)_A - t_M}$
Resolution	$R_S = \frac{2[(t_R)_B - (t_R)_A]}{W_A + W_B}$
Number of plates	$N = 16\left(\frac{t_R}{W}\right)^2$
Plate height	$H = L/N$
Retention time	$(t_R)_B = \frac{16 R_S^2 H}{u}\left(\frac{\alpha}{\alpha - 1}\right)^2 \frac{(1 + k'_B)^3}{(k'_B)^2}$

(45pts) 2. Two weak acids, HA and HB have K_a values of 4×10^{-7} and 5×10^{-4}, respectively. Their distribution coefficients (K_D) between water and chloroform are 5.5 for HA and 4.5 for HB. Design a procedure to separate these two acids by extracting 98% of the HA from aqueous solution into the chloroform, while at the same time retaining HB in the aqueous phase with no more than 1% loss (i.e. contamination) with each extraction.